SM2501/FSG3084

Lecture 15

Weakness and compensation

Walking aids

Performance enhancing devices





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1

Muscle weakness causes

Neuromuscular pathology

- Muscular dystrophy, etc
- Pathology of neuromuscular junction

Neurological pathology or injury

- Cerebral palsy, stroke
- Myelomeningocele, etc

Disuse

- Consequence of deformity (arthrygryposis, etc)
- Consequence of pain (rheumatoid arthritis, etc)

Fatigue (temporary)

2

Management

Strength training and physical therapy

Pharmaceutical treatment

Bracing and/or walking aids

Realistic goals

3

Measurement of strength

Functional tasks

- Number of heel rises while standing, etc.

Manual muscle testing

- Grade 0: No contraction or muscle movement
- Grade 1: Trace of contraction, but no movement at the joint
- Grade 2: Movement at the joint with gravity eliminated
- Grade 3: Movement against gravity, but not against added resistance
- Grade 4: Movement against external resistance, but less than normal
- Grade 5: Normal strength

Dynamometry

- Isometric
  - Handheld
  - Instrumented







Moment arm measured

4

Instrumented isometric



SMC Karolinska Institutet
Örtqvist et al 2007

Somewhere in between…



Instrumented isokinetic





5

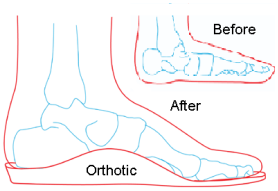
Use of orthoses

Purposes

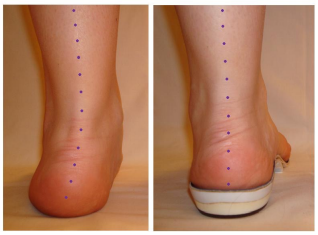
- Maintain or correct body segment alignment
- Assist joint or resist joint motion during key phases of gait
- Relieve or distribute distal weight bearing forces
- Protect and support against injury
- Increase mobility
- Minimize risk of deformities

6

Foot orthoses (FO)







Goals: often to correct foot and ankle alignment, which also can correct proximal loading

7

Supramalleolar orthoses (SMO)





For significant forefoot, midfoot and subtalar instability

8

Ankle-Foot Orthoses (AFO)




GRAFO (resists dorsiflexion)
&
Posterior leaf spring (resists plantarflexion, allows some dorsiflexion)




Hinged, with plantarflexion stop

Allows dorsiflexion, prevents plantarflexion, depending on physical constraints



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Solid or near solid

Adjustable neutral angle

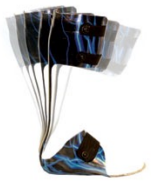
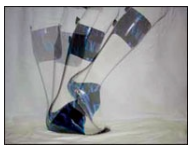
Continued

9

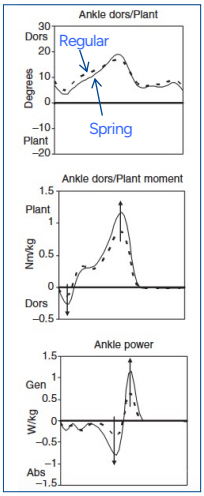
Carbon fiber spring ankle joint
(polycentric joint)




Technology similar to prosthetic 'blades'

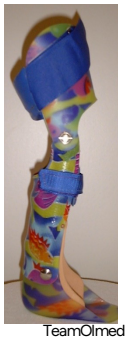
Store energy in tibial advancement
Return in pre-swing



Bartonek et al 2007

10

Knee-Ankle-Foot Orthoses (KAFO)

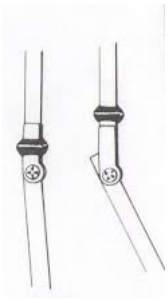


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Open hinge joint

Aligns in frontal and transverse plans





Locking knee joint

Opens for sitting

Can be combined with different ankle joint types
Have been some attempts to lock knee during stance, unlock during swing

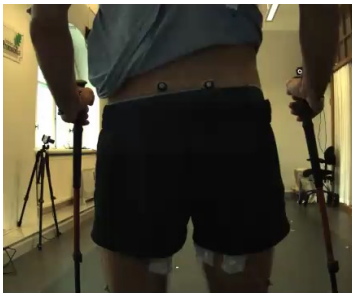
11

Incomplete spinal cord injury, partial and asymmetric weakness





Rigid AFOs



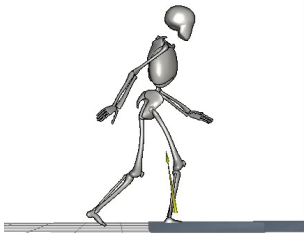
Rigid AFOs + walking sticks

12

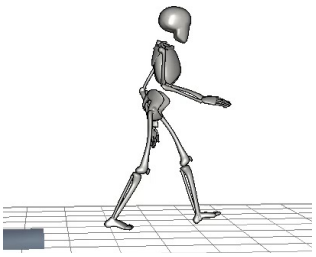
12

MMC: Plantarflexion paresis, reduced strength in hip abductors (n=6)

10 y, barefoot



Solid AFOs



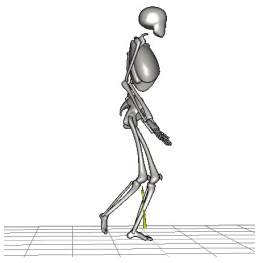
↑ trunk sway (Δ≈4°)
↑ hip extension (Δ≈6°)
↑ knee ROM (Δ≈16°)
↑ knee extension in midstance (Δ≈15°)
↓ dorsiflexion (Δ≈12°)
↑ plantarflexion (Δ≈6°)
↑ plantarflexor moment (Δ≈70%)
longer steps, lower cadence

Gutierrez et al, 2005

13

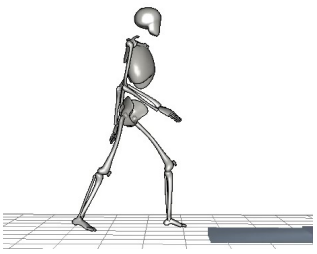
MMC: Paresis of plantarflexors, dorsiflexors, hip abductors. Partial paresis of hip extensors and knee flexors (n=8)

13 y, barefoot



Risk of knee 'valgus thrust'

Solid AFOs



↑ hip extension (Δ≈6°)
↑ hip ROM (Δ≈6°)
↑ hip flex. mom. (Δ≈18%)
↑ knee ROM (Δ≈19°)
↓ dorsiflexion (Δ≈12°)
↓ external foot progression (Δ≈6°)
↑ plantarflexor moment (Δ≈135%)
longer steps, lower cadence

Gutierrez et al, 2005

14

(Knee Orthoses (KO))



Össur

Unloader braces (3-4 point bending)

Unload medial or lateral compartment
Osteoarthritis



Immobilizing and supporting braces

Restricts motion
Ligament or cartilage tears, trauma

And probably hundreds more.....

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Hip-Knee-Ankle-Foot Orthoses (HKAFO) Recriprocating gait orthosis (RGO)



HKAFO

More extensive paralysis

RGO



Hip flexion through posterior trunk lean

https://www.youtube.com/watch?v=IgTZXCLNhs4

https://www.youtube.com/watch?v=cPjA8JJe7YU

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Standing Orthoses/Frames

Thoracic-Hip-Knee-Ankle-Foot-Orthosis (THKAFO)

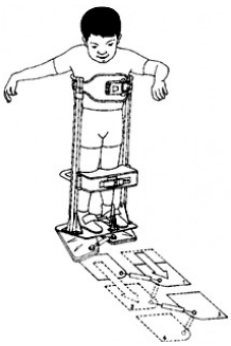

Standing Frame



For severe paralysis.
Prevents contractions, loads skeleton, aids in circulation and digestion.
Can be used in young children before RGO.

17

Swivel Walker ('Standing Walker')





Also for severe paralysis.
Allows some upright mobility and household ambulation.

https://www.youtube.com/watch?v=jyfvgflUys8

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Prostheses

Unpowered/passive

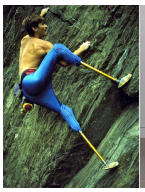


Jointed



Polycentric joint

Powered/active





Hugh Herr: The new bionics that let us run, climb and dance | TED Talk

Hugh Herr: How we'll become cyborgs and extend human potential | TED Talk



Össur Symbionic



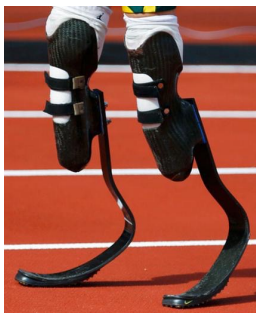
Ottobock C-leg

19

Performance enhancement in sport - "Technical doping"?



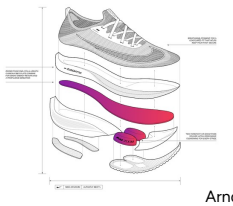
LZR Racer, Speedo, "Sharkskin"

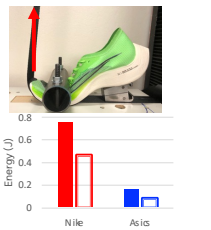


Carbon fiber prostheses

https://www.facebook.com/watch/?v=2884731111773548







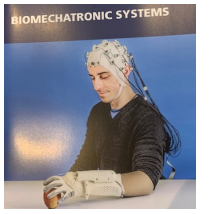
Arndt, Hallström, Gutierrez-Farewik

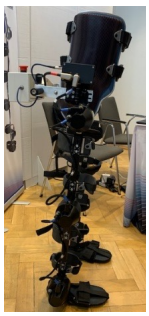
Carbon fiber plates embedded in shoe foam midsole
(Nike Alphafly, etc.)

20

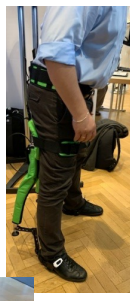
Exoskeletons

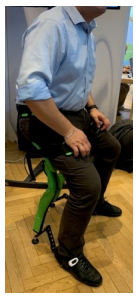
A number of exoskeleton prototypes and products

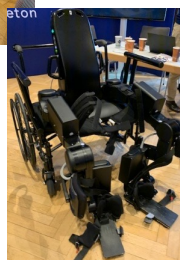














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Strength/performance-enhancing, injury risk-reducing



Raytheon XOS 2



BLEEX



AIRFRAME, Levitate Technologies

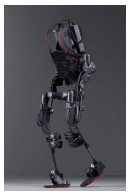


RoboGlove, Bioservo

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Better known active and passive exoskeletons

Commercially-available



Ekso Bionics



HAL, Cyberdyne



NASA X1 Mina

Research

Active



Soft Exosuit, Harvard
(Walsh et al)





MIT Biomechatronics
(Herr et al)

Passive





Stanford Biomechatronics
(Collins et al)



Georgia Tech
(Sawicki et al)

23

Exoskeletons/wearable robotics

Autonomous



Locomat

Commonly after CNS trauma
Rehabilitation & gait training/retraining

https://www.youtube.com/watch?v=vc7nGsS3VVY

EMG-driven, EEG-driven



Zhang, Gutierrez-Farewik



Goordleva et al. 2020

Development/research phase mostly

Semi-autonomous



Ekso
Bionics

https://www.youtube.com/watch?v=9o9LdRlikGQ&t=70s



ABLE



TWIICE

Hybrid



HAL Cyberdyne

EMG <-> autonomous

Rehabilitation and gait retraining after neurotrauma, gait assistance

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Human-in-the-loop optimization

Multi-objective, for assistance of dropfoot gait after stroke



Unpowered



Powered, optimized via HILO

Zhang, Gutierrez-Farewik

Tethered emulators

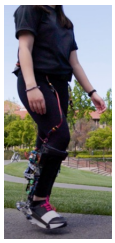


Stanford and Carnegie Mellon
https://biomechatronics.stanford.edu/human-loop-videos



Harvard
https://www.facebook.com/watch/?v=10155425285613519

Outside of lab



Stanford
https://www.youtube.com/watch?v=-vTk-TRN0C8

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Challenges

- Simulation can save time and interations in prototyping and experiments, but
  - People are not always predictable
  - Human bodies are not rigid
  - Translating an ideal actuator to an actual device is not trivial
- Batteries are heavy and inefficient
- Motors are often noisy
- Pneumatic devices are slow and require compressed air
- Detecting and predicting intended motions is difficult
  - At all
  - In time to inform and adjust a controller
- A device necessarily applies interaction forces on the user. Comfort?
- One size doesn't fit all
- …

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Next 5-10 years?

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